

Cryptography - 10_10

Next question is: How do we build $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1}$?

- Practice: Heuristic construction.
- Theory: Provably from any OWF.

In theory we can just believe OWF and build G from it.

In order to do this, we need a fundamental concept: Hard-core bits. They are bits of info about x that are hard to compute given $y = f(x)$.

I have a OWF, I pick a random x , and give you y . Because it's OWF, we cannot compute x from y , but there is a specific bit of x that's hard to compute. Usually it doesn't need to be a specific bit, instead a predicate $h(x)$, such that $h(x) \in \{0,1\}$ is hard to compute given $f(x)$ (with probability better than $\frac{1}{2}$).

Why is this relevant? Assume that $h(x)$ exists, and it's hard to predict with a probability better than $\frac{1}{2}$, then it means that it's random.

First question: Can there be a single h such that h is hard-core $\forall f$ (OWF)? No, it's impossible. Provide $\hat{f}$ (OWF) such that h not hard-core for $\hat{f}$. Fix any h : Take f (OWF), then consider $\hat{f}(x) = h(x) || f(x)$ ($h(x)$ concatenated with $f(x)$). h not hard-core for $\hat{f}$. In fact, is $\hat{f}$ still OWF?

If it's not one way, there is an adversary $\hat{\mathcal{A}}$ able to compute the inverse. Then we can build $\mathcal{A}$ to invert f as well, which we defined as a OWF (? , check this). $\mathcal{A}$ talks with $\hat{\mathcal{A}}$ giving him a guessed bit b followed by y , given by the challenger. Then:

$$Pr[\hat{\mathcal{A}} \text{ wins}] \geq \frac{1}{2} \cdot Pr[\hat{\mathcal{A}}(h(x) || y) \text{ wins}]$$

$$\begin{aligned} Pr[\hat{\mathcal{A}} \text{ wins}] &= Pr[\hat{\mathcal{A}}(b, y) \text{ wins} \wedge b = h(x)] + Pr[\hat{\mathcal{A}}(b, y) \text{ wins} \wedge b \neq h(x)] \geq \\ &\geq \frac{1}{2} \cdot Pr[\hat{\mathcal{A}}(h(x), y) \text{ wins}] \geq \frac{1}{2} \cdot \frac{1}{poly} \geq \frac{1}{poly} \end{aligned}$$

(then not negligible).

We can, then, swap the quantifies, such that $\forall f \exists h$.

✓ Definition

Let $f : \{0,1\}^n \rightarrow \{0,1\}^n$ a OWF. Then h is hard-core for f if:

1. $\forall PPT \mathcal{P} \Pr[\mathcal{P}(y) = h(x)]$ with $x \in \{0, 1\}^n$ and $y = f(x)$ is $\leq \frac{1}{2} + \text{negl}(n)$
2. (Alternative) $(f(x), h(x)) \approx_c (f(x), b)$, with $x \in \{0, 1\}^n$ and $b \in \{0, 1\}$

🔗 Exercise

2 $\implies$ 1. Assume 1 does not hold, then I build a $\mathcal{P} \dots$

Also true that 1 $\implies$ 2.

✓ Theorem

If one-way permutations (OWF) exist $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, then $\exists G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ a PRG.

📖 Proof

We define $G(s) = f(s) || h(s)$ (with s being the seed).

We have that $G(U_n) \equiv f(U_n) || h(U_n) \approx_c f(U_n) || U_1 \equiv U_{n+1}$

■

✓ Theorem

If OWF exist, also PRG with $l(n) = 1$ exist.

All is left is to build h for any given f .

✓ Theorem (Goldreich-Lenin)

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a OWF. Then, $g(x, r) = (f(x), r)$ is also OWF. So $g : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ is also OWF with hard-core predicate

$$h(x, r) = \bigoplus_{i=1}^n x_i \cdot r_i = \langle \vec{x}, \vec{r} \rangle \pmod{2}$$

Proof ideas: If $\exists$ PPT $\mathcal{P}$ for $h(x, r)$, then $\exists$ PPT $\mathcal{A}$ breaking g , in particular $\mathcal{A}$ can find x .

Simple cases:

- Assume $\mathcal{P}$ is super good, $\forall x, r \Pr[\mathcal{P}(y) = h(x, r)] = 1$

Then $\mathcal{A}$ will just run $\mathcal{P}$ on $y_1 = (f(x), \vec{e_1})$, then $y_2 = (f(x), \vec{e_2})$ and so on, with e_i the vector with all zeros except for the position i , where he has 1.

- Second idea: Assume that $\mathcal{P}$ is very good, $\forall x \in \{0, 1\}^n$

$$\Pr[\mathcal{P}(f(x), r) = h(x, r)] \geq \frac{3}{4} + \frac{1}{poly}.$$

The idea is that we run $\mathcal{P}$ on r random and $r \oplus e_i$.

We'll still have that $x_i = \langle x, r \oplus e_i \rangle \oplus \langle x, r \rangle = \langle x, e_i \rangle$

Still, you can amplify by taking majority of many queries.